Particulate Organic Carbon/Nitrogen Filtering Standard Operating Procedure

Version 1.0

Authors

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Protocol Revision Record

Version	Date Implemented	Description of Revisions
1.0	10 May 2025	Initial release in format standardized for Bio-GO-SHIP

Acronyms, Abbreviations, Terminology

Term	Definition
Inline/underway water	Sampling water that is pumped from beneath the ship to a receiving spigot in an onboard laboratory
CTD rosette water	Sampling water gathered from Niskin bottles on the conductivity, temperature, density (CTD) rosette
DI water	Deionized water from the Milli-Q system
POC/N	Particulate organic carbon/nitrogen
GF/F	Glass fiber filter
GO-SHIP	Global Ocean Ship-Based Hydrographic Investigations Program
Bio-GO-SHIP	Biological GO-SHIP

Background

Summary

This protocol describes discrete sampling of small volume (0.5-2 L) POC/N while onboard a research vessel, with sampling water from the CTD rosette or underway system.

Method description and rationale

POC/N will be determined for samples collected from both inline flow through systems and CTD bottle rosettes. For POC/N samples, a single volume ranging from 0.5 to 3 L depending on particulate load will be collected onto pre-combusted (450C, 4 hours) 25 mm Whatman GF/F filters via low pressure (less than 5 mm Hg) vacuum filtration. Filters will be folded in half, sample side inward, and wrapped in pre-combusted foil packets (450C, 4 h) and stored in LN until sample preparation and analysis. POC values must be corrected for non-target carbon from filter blanks and DOC adsorption. As such, the regression approach consisting of three volumes (typically 0.5 L, 1 L, and 2 L) will be performed once daily either in the surface mixed layer or from samples collected below the mixed layer. Both are required to constrain blanks for samples collected as part of a vertical profile. Two estimates of filter blanks are produced by the regression method. The blanks include (1) a 1 L sample of filtered seawater retained from each 1 L filtration and then re-filtered through a new pre-combusted GF/F and (2) an intercept blank from the regression of carbon values and volumes for three POC/N samples. Duplicate samples will be collected at a rate of one every ten samples. NASA Goddard Space Flight Center Ocean Ecology Laboratory sample processing methods include an assessment of the duplicate samples, sample precision by reinjection of samples, and an effective limit of quantitation (lower detection limit) which are all provided with the data report.

Spatial coverage and environment(s) of relevance

This protocol can be used across any marine environment to effectively collect POC/N data from surface waters and at depth.

Ship space requirements

About 4 square feet of counter space is needed for filtration rigs and vacuum pump, and a waste carboy will be secured nearby on the ground.

Personnel Required

One person that is trained in filtration protocol. Typically, two individuals that operate on 12-hour shifts.

Safety

There are no major safety concerns with this protocol.

Time needed to execute the procedure

To complete protocol from gathering sample water, filtering, and storing samples may take about 30 minutes.

Standard Operating Procedure

Materials:

- 25 mm 0.7 um GF/F filters, pre-combusted at 450 C for 4 hours
- 2L dark sample bottles
- Combusted foil envelopes

Filtering instructions (3 volume regression):

- Wear gloves when gathering sample water and filtering
- Prep filter frit ahead of time by using forceps to place one combusted GF/F on the frit, reattach filter cup, dampen GF/F with DI water, and cover filter cup with a piece of foil
    - One filter frit should be the glass rig.
    - The regression volumes are 0.5 L, 1 L, and 2.xx L (volume of 2 L is based on what is written on sample bottle). For the 1 L sample, use the 1 L line on the 2 L bottles. For the 0.5 L sample, use the 0.5 L line on the 1.0 L sample bottle
- Rinse sample bottle 3 x with sample water
- Invert sample bottle over the filter frit and GF/F
- Filter sample in relatively dark room or space, e.g. overhead lights off or covered
- Watch water carefully – let the filter just barely run dry as you stop filtering
- Use forceps to fold GF/F in thirds or quarters, sample side folded inwards. avoid touching the sample area
- Store filter in a labeled combusted foil envelope
    - Label as: POC, date, station number, volume filtered, initials
    - Include A, B, or C if there are replicates at the same station
- Place a blank filter on a new frit, and use the water in the beaker to filter a second time as a “wet blank”

- Include wet blank with every small volume POC sample, include a dry blank roughly every 3 days
- Place foil packets into the -80 or liquid nitrogen. All packets can go into the same Ziploc bag and they will get sorted out later in the lab
- Empty waste carboy
- Fill out elogs

Basic troubleshooting

Contact Jason Graff via WhatsApp for any issues encountered.